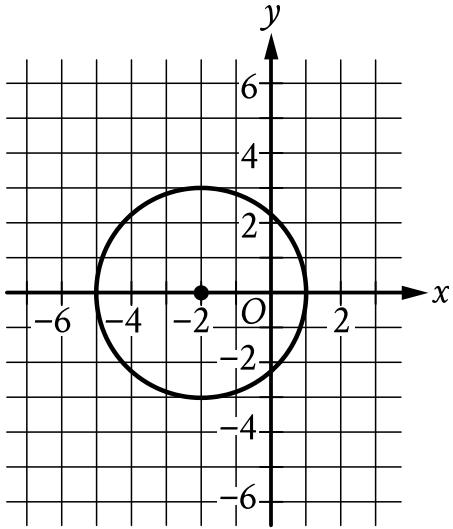


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Circle A (shown) is defined by the equation $(x + 2)^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. Which equation defines circle B?

- Answer
- A. $(x + 2)^2 + (y + 6)^2 = (4)(9)$
 - B. $2(x + 2)^2 + 2(y + 6)^2 = 9$
 - C. $(x + 2)^2 + (y - 6)^2 = (4)(9)$
 - D. $2(x + 2)^2 + 2(y - 6)^2 = 9$

Correct Answer: A

Rationale

Choice A is correct. According to the graph, the center of circle A has coordinates $(-2, 0)$, and the radius of circle A is 3. It's given that circle B is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. It follows that the center of circle B is 6 units below the center of circle A. The point that's 6 units below $(-2, 0)$ has the same x-coordinate as $(-2, 0)$ and has a y-coordinate that is 6 less than the y-coordinate of $(-2, 0)$. Therefore, the coordinates of the center of circle B are $(-2, 0 - 6)$, or $(-2, -6)$. Since the radius of circle B is 2 times the radius of circle A, the radius of circle B is $(2)(3)$. A circle in the xy-plane can be defined by an equation of the form $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the radius of the circle is r . Substituting -2 for h , -6 for k , and $(2)(3)$ for r in this equation yields $(x - (-2))^2 + (y - (-6))^2 = ((2)(3))^2$, which is equivalent to $(x + 2)^2 + (y + 6)^2 = (2)^2(3)^2$, or $(x + 2)^2 + (y + 6)^2 = (4)(9)$. Therefore, the equation $(x + 2)^2 + (y + 6)^2 = (4)(9)$ defines circle B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation defines a circle that's the result of shifting circle A up, not down, by 6 units and increasing the radius.

Choice D is incorrect and may result from conceptual or calculation errors.